

Guidelines



Research Proposal



The Research



Deadline

**Thesis Report and Artefact Submission – 28th
August 2026**



海未

@annafleur



My university motto:

"If tomorrow isn't the due date,
today isn't the do date"

What posters do

- Is a BIG piece of paper, that can be use to explain your research.
- It should attract viewers.
- It should demonstrate your knowledge and skills.
- Different purposes need different types of poster – for instance, a conference poster is very different from an advertising poster
- Main audience of your poster would be
 - Specialist in the research area
 - Other students
 - Staff



Introduction

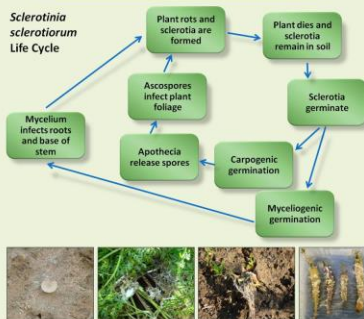
The ascomycete fungus *Sclerotinia sclerotiorum* is a necrotrophic pathogen with a host range of over 400 plant species including economically important crops such as lettuce, vegetable brassicas, oilseed rape, beans, peas, potatoes and carrots. UK carrot growers suffer estimated annual crop losses in excess of six million pounds due to the disease, with marketable yield estimated to be reduced by 1 tonne per hectare for each 1% increase in diseased roots. The pathogen survives in the soil as sclerotia (resting bodies which can survive for at least 5 years) which germinate when brought close to the soil surface to produce mushroom-like apothecia. These then release air-borne ascospores which infect plants and eventually further sclerotia are formed on diseased plants and are returned to the soil. *S. sclerotiorum* is a diverse pathogen with multiple genotypes being found within a single crop in the UK. A relative of *S. sclerotiorum*, *S. subarctica* has been recently identified in the UK after previously only being found in Norway on wild hosts and on vegetable crops in Alaska. The symptoms caused by *S. subarctica* are very similar to *S. sclerotiorum* and therefore may be undetected in crops in the UK. Control methods for Sclerotinia disease are predominantly fungicides and cultural practices such as foliage clipping or crop rotations. Products such as Contans WG (biocontrol agent) and Perika (Calcium cyanamide) provide long-term control against the disease by reducing the numbers of sclerotia in the soil, but are generally considered too costly by growers.

Aims

To identify potential new soil treatments for control of Sclerotinia disease and to assess the impact of pathogen diversity on both aggressiveness and fungicide sensitivity.

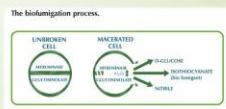
Objectives

1. To determine the effect of organic soil amendments on the survival of sclerotia of *Sclerotinia sclerotiorum*.
2. To determine the aggressiveness of different *Sclerotinia* genotypes and species on commercial carrot varieties and quantify production of sclerotia.
3. To evaluate the sensitivity of different *Sclerotinia* genotypes and species to fungicides.
4. To investigate the epidemiology and control of *Sclerotinia subarctica*.
5. To carry out a population study of *S. sclerotiorum* on *Daucus carota* in the UK.



Biofumigation

Using Brassica green manure crops for biofumigation can provide control against Sclerotinia disease, but has not yet been shown to have a consistent significant effect on viability of sclerotia. Many Brassica species produce significant levels of glucosinolates, which are hydrolysed in the presence of water and endogenous myrosinase enzyme to release isothiocyanates (ITCs) which have a wide range of biocidal characteristics.



Biofumigation Trial – Soil Box Method

Field rate equivalent of treatments (plant material macerated first) were mixed with pasteurised compost and this compost mixture was placed into 600ml hotform boxes with 30 pre conditioned sclerotia arranged in a grid pattern. The sclerotia were covered with more of the compost mixture, water was added and lids were immediately placed on the boxes. Boxes were incubated at 15°C and checked for germination twice a week.

4 replicates of the following treatments:

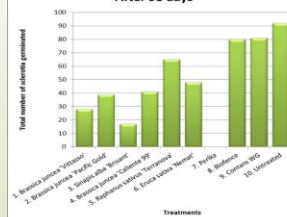
1. *Brassica juncea* 'Vittasso'
2. *Brassica juncea* 'Pacific Gold'
3. *Sinapis alba* 'Brisant'
4. *Brassica juncea* 'Caliente 99'
5. *Raphanus sativus* 'Teranova'
6. *Eruca sativa* 'Nemat'
7. Biofence
8. Perika (calcium cyanamide)
9. Contans WG (*Coniothyrium minitans*)
10. Untreated



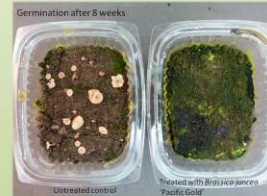
Results From Initial Soil Box Trial

Carpogenic germination has been affected by the biofumigant crops, with white mustard treatment (*Sinapis alba* 'Brisant') showing the greatest reduction in germination after 98 days.

Soil Box Trial 1 - Total Germination After 98 days

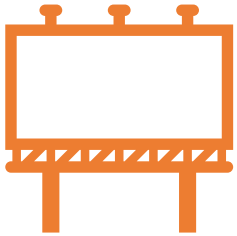


All of the biofumigation crop treatments delayed carpogenic germination of sclerotia in comparison to the untreated control.



Conference
Poster

Key concepts



The audience are interested in what your posters *do*, not how pretty they are



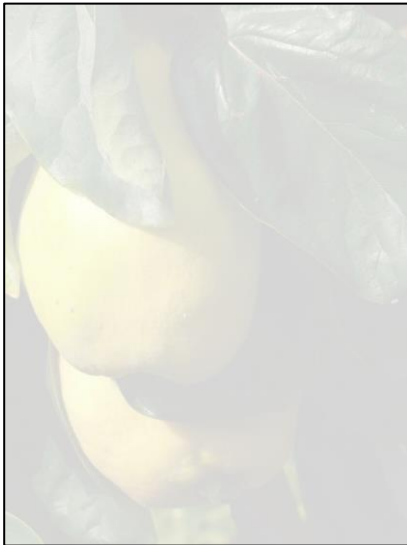
Your poster needs to attract people from three distances.

First distance: Across the room. Is your poster interesting enough to make someone cross a room to see more detail?

Second distance: A couple of metres away. When someone can see the subtitles and images on your poster, do they want to come closer?

Third distance: Close enough to read everything. Now that they can see everything, do they want to read everything?

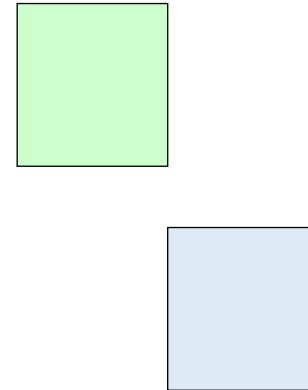
Some ways of catching attention from across a room



Remember: The image should be relevant. This poster might be for a project about IT support for fruit farmers.

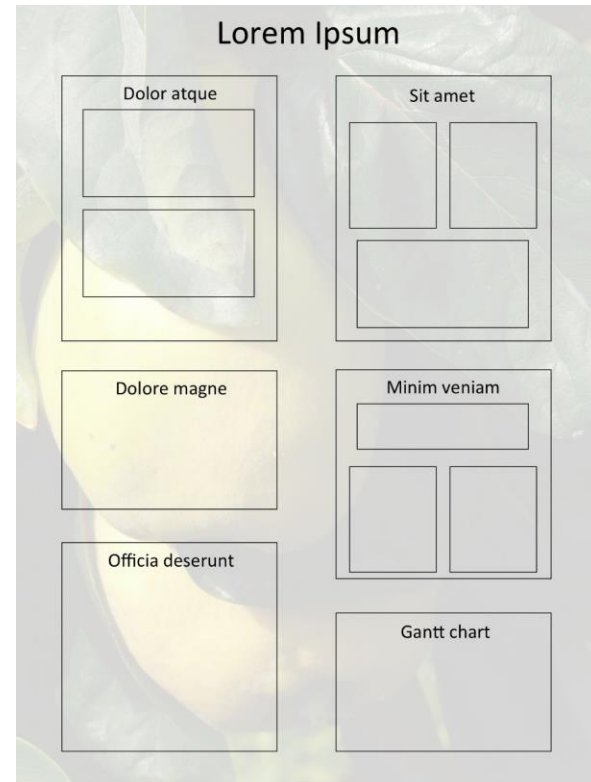
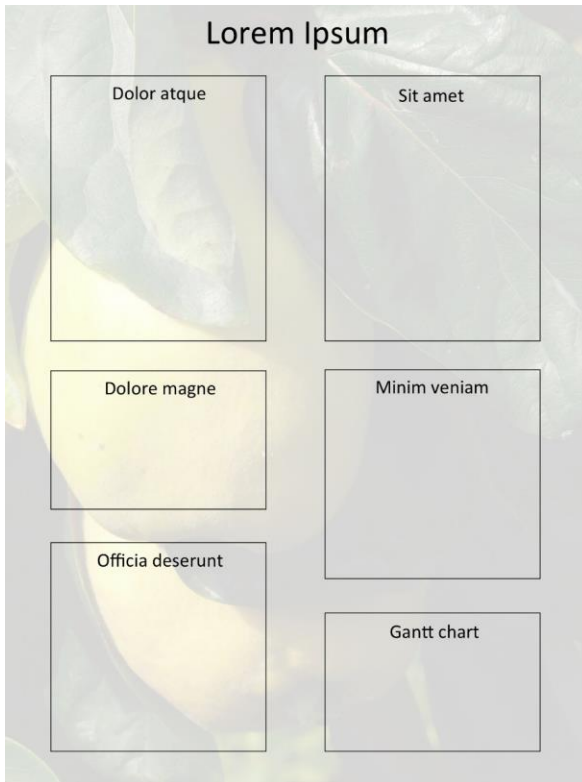
Excellent title

Remember: Check with your supervisor about whether it really is an excellent title.



Remember: Make sure that the images show skill and knowledge from the course.

Examples of structure and layout



Layouts

[illegible][illegible]

Keeping the viewer interested at close range



At this range, a background image can be a distraction – if you use one, make sure it's not too obtrusive at close range (try viewing the poster on screen at 100% magnification)



Pay a lot of attention to detail in formatting (e.g. aligning the pictures)



Pay a lot of attention to detail in the content (e.g. spell checking; showing knowledge of technical and academic content from the course)



Show the viewer interesting information about your project; don't just tell them things they know already



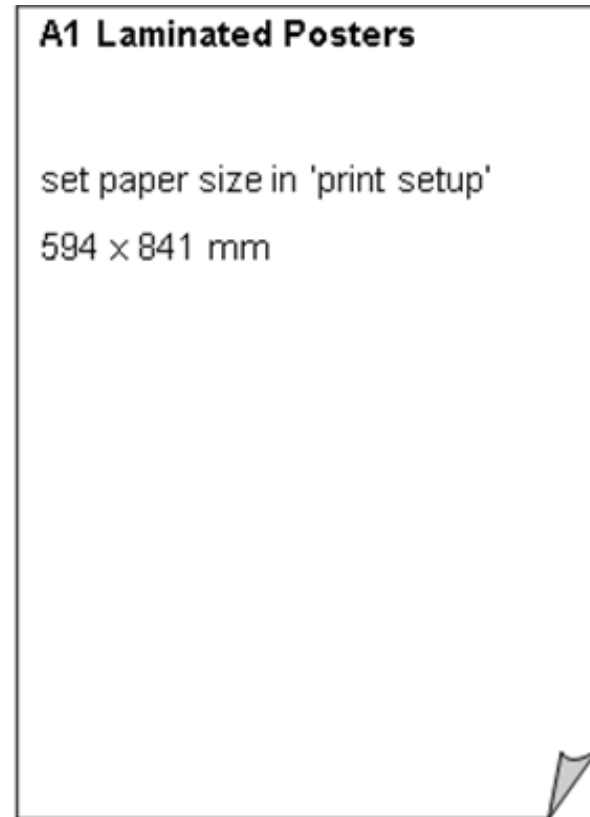
Remember to include some references to the literature, to show your background reading



Use different text sizes for different purposes

Structure and layout

- Your poster should be A1 in size.
 - Portrait Layout



Example - Bad Design




PIGS IN SPACE: EFFECT OF ZERO GRAVITY AND AD LIBITUM FEEDING ON WEIGHT GAIN IN CAVIA PORCELLUS

Colin B. Purrington
6673 College Avenue, Swarthmore, PA 19081 USA




ABSTRACT:

One ignored benefit of space travel is a potential elimination of obesity, a chronic problem for a growing majority in many parts of the world. In theory, when an individual is in a condition of zero gravity, weight is eliminated. Indeed, in space one could conceivably follow as libitum feeding and never even gain an gram, and the only side effect would be the need to upgrade one's stretchy pants ("exercise pants"). But because many diet schemes start as very good theories only to be found to be rather harmful, we tested our predictions with a long-term experiment in a colony of Guinea pigs (*Cavia porcellus*) maintained on the International Space Station. Individuals were housed separately and given unlimited amounts of high-calorie food pellets. Fresh fruits and vegetables were not available in space so were not offered. Every 30 days, each Guinea pig was weighed. After 5 years, we found that individuals, on average, weighed nothing. In addition to weighing nothing, no weight appeared to be gained over the duration of the protocol. If space continues to be gravity-free, and we believe that assumption is sound, we believe that sending the overweight — and those at risk for overweight — to space would be a lasting cure.

INTRODUCTION:

The current obesity epidemic started in the early 1990s with the invention and proliferation of elastane and related stretchy fibers, which released wearers from the rigid constraints of clothes and permitted monthly weight gain without the need to buy new outfits. Indeed, exercise today for hundreds of millions of people involve only the act of wearing stretchy pants in public, presumably because the constrictive pressure forces fat molecules to adopt a more compact tertiary structure (Xavier 1965).

Luckily, at the same time that fabrics became stretchy, the race to the moon between the United States and Russia yielded a useful fact: gravity in outer space is minimal to nonexistent. When gravity is zero, objects cease to have weight! Indeed, early astronauts and cosmonauts had to secure themselves to their ships with seat belts and sticky boots. The potential application to weight loss was noted immediately, but at the time travel to space was prohibitively expensive and thus the issue was not seriously pursued. Now, however, multiple companies are developing cheap extra-orbital travel options for normal consumers, and potential travelers are also creating new ways to pay for products and services that they cannot actually afford. Together, these factors open the possibility that moving to space could cure overweight syndrome quickly and permanently for a large number of humans.

We studied this potential by following weight gain in Guinea pigs, known on Earth as fond of ad libitum feeding. Guinea pigs were long envisioned to be the "Guinea pigs" of space research, too, so they seemed like the obvious choice. Studies on humans are of course desirable, but we feel this current study will be critical in acquiring the attention of granting agencies.

MATERIALS AND METHODS:

One hundred male and one hundred female Guinea pigs (*Cavia porcellus*) were transported to the International Space Laboratory in 2010. Each pig was housed separately and deprived of exercise wheels and fresh fruits and vegetables for 48 months. Each month, pigs were individually weighed by duct-taping them to an electronic balance sensitive to 0.0001 grams. Back on Earth, an identical cohort was similarly maintained and weighed. Data was analyzed by statistics.



CONCLUSIONS:

Our view that weight and weight gain would be zero in space was confirmed. Although we have not replicated this experiment on larger animals or primates, we are confident that our result would be mirrored in other model organisms. We are currently in the process of obtaining necessary human trial permissions, and should have our planned experiment initiated within 80 years, pending expedited review by local and Federal IRBs.

ACKNOWLEDGEMENTS:

I am grateful for generous support from the National Research Foundation, Black Hole Diet Plans, and the High Fructose Sugar Association. Transport flights were funded by SPACE-EXES, the consortium of wives divorced from insanely wealthy space-flight startups. I am also grateful for comments on early drafts by Mariana Athletic Club, Corpus Christi, USA. Finally, sincere thanks to the Guy Foundation for generously donating animal care after the conclusion of the study.

RESULTS:

Mean weight of pigs in space was 0.0000 ± 0.0002 g. Some individuals weighed less than zero, some more, but these variations were due to reaction to the duct tape, we believe, which caused them to be alarmed push briefly against the force plate in the balance. Individuals on the Earth, the control cohort, gained about 240 g/month ($p = 0.0002$). Males and females gained a similar amount of weight on Earth (no main effect of sex), and size at any point during the study was related to starting size (which was used as a covariate in the ANCOVA). Both Earth and space pigs developed substantial dewlaps (double chins) and were lethargic at the conclusion of the study.

LITERATURE CITED:

NASA. 1962. Project STS-XX: Guinea Pigs. Leaked internal memo.

Sekiuti, S.R., D. D. Lukac, and N.M. Neumov. 2005. The Fetus Cannot Exercise Like An Astronaut: Gravity Loading Is Necessary For The Physiological Development During Second Half Of Pregnancy. Medical Hypotheses. 64:221-228

Xavier, M. 1965. Elastane Purchases Accelerate Weight Gain In Case-control Study. Journal of Obesity. 2:23-40.

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<http://colinpurrington.com/tips/academic/posterdesign>

Poster Content



Title of project; student name; supervisor name



Project description; aims and objectives



List of requirements and/or brief list of relevant background literature



Methodology



Results, Screen shots of the interfaces



Sample of annotated code / flow charts or tabulated raw data or other results



What has been achieved and what is yet to be done



Reference

Software to
be used

Microsoft PowerPoint

Visio

Illustrator

Canva

Photoshop

Convert the document to PDF or Jpeg image
before you take the printout or submit online

Final Report Structure



An illustration of a mother and daughter reading a book together at night. The mother, wearing glasses and a pink shirt, is holding the book. The daughter, wearing a yellow shirt, is looking at the book. A teddy bear is sitting on the bed next to them. The background is a dark purple night sky with yellow stars and a yellow moon. A green bird is flying in the sky. The scene is framed by a white, stylized tree branch with green leaves.

Once Upon A Time

A Collection of Bedtime Stories

Written by

Marthina Smith

Telling a
story



Telling a Story

- The report, in effect, tells the story of your project
- If you are telling a story you have a beginning, a middle and an end
 - in the beginning, you introduce the characters in the story and give some information about them – where they live, how old they are, etc.
 - in the middle, you relate what happened, how the characters reacted, etc.
 - finally, you tell how everything was resolved

Writing a Report

- This doesn't mean you tell it as if it's a diary
 - and then I did this and then I did that
- There are a couple of minor additions - people often have to search through reports to find the ones that interest them, so to help this, you also provide brief summaries of:
 - what the report contains (abstract)
 - what the main outcomes were and what the implications of these results are (conclusion)





Report Format

- Follow formatting guidelines.

Final Report Structure

- Title page(1 A4 page) - see the sample
- Declaration (1 A4 page)
- Abstract (1 A4 page)
- Acknowledgement (1 A4 page)
- Table of contents (1-2 A4 pages)
- List of Tables (if any)
- List of Figures (if any)
- List of abbreviations (if any)

- One para
 - NO references / no foot notes
 - 250 – 350 words
 - This is about PROJECT !! Not report
-
- What is your project!!!
 - Why ? (is it needed) – current situation
 - Uniqueness
 - How ??
 - Benefits / Services / end results



Final Report Structure

- Introduction (5-7 A4 pages)
 - Background context
 - Research problem to be addressed
 - Research questions
 - Research objectives
 - Scope of the research
 - Research Contribution
 - Structure of the report
- Literature Review(8– 10 A4 pages)
 - Introduction
 - Sub headings
 - Research Gap



Final Report Structure

- **Methodology / Results / Discussion**
- Conclusion (2-4 A4 pages)
- References (no limit)
- Glossary (if any-1-2 pages)
- Appendices (no limit)

Methodology / Results
/Discussion
– System Development Projects



Methodology

– Things you should include

Explain in detail about the methodology followed to make the research complete.

How was project planned?

How was data gathered and analyzed?

The project design

- Software Architecture diagram - (diagram + explanation)
- Class Diagram / ER Diagram - - (diagram + explanation)
- Interfaces / Wireframe - (diagrams + explanation)
- Circuit Diagram - (diagrams + explanation)
- Model Design- (diagrams + explanation)
- Algorithm Design- (flow chart + explanation)

Methodology

– Things you should include

How system was implemented.

Hardware / Software used /
Reasons to select those

Results

- Explain based on modules / major functions.
- Screenshots / Pseudocode / Code fragments / Flowcharts with detail description.
- System Test Cases for all the main functionality of your project.
- Test case format with actual results

4. Results

- 4.1 Data Model Development

- 4.1.1 Model for the face recognition

- What it is

- How you developed , tools used

- Interfaces – explanations

- Code segments

- Flow charts

- Test cases

- 4.1.2. Model to detect the emotion

- What it is

- How you developed , tools used

- Interfaces – explanations

- Code segments

- Flow charts

- Test cases

- 4.2 Mobile Application Development

- 4.3 API Development

- 4.4 System Integration

Methodology / Results
/Discussion
– *Concept Projects*

Methodology – Things you should include

- Study type (Qualitative / Quantitative / Mix)
- Study Design
 - Theoretical Framework
- Data collection methods
- Data analysis methods
- Results
 - Statistics
 - Graphs
 - Tables
 - Tables as appropriate
- Discussion
 - What did you find?
 - What does it tell us?
 - How does it relate to the literature and what we were expecting?

Example of a code segment

```
import java.io.*;

public class Account {

    public static void main (String args[]) throws
IOException
    {
        int accNo; String name, type, s; double balance;

        System.out.println("Account Name: " + name +
"\n" + "Account number : " + accNo + "\n" + "Account
type: " + type + "\n" + "Account Balance: " + balance);
    }
}
```

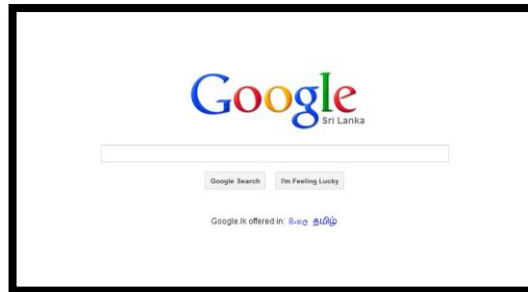
Listing 1 : Account class

Sample Test case

Test Case : Admin Login**Pre Conditions : The admin wants to login to the system**

TC #	Sequence #	Test Description	Input Value	Expected Result
1	1.1	Description	Value	Pass / Fail
	1.2			

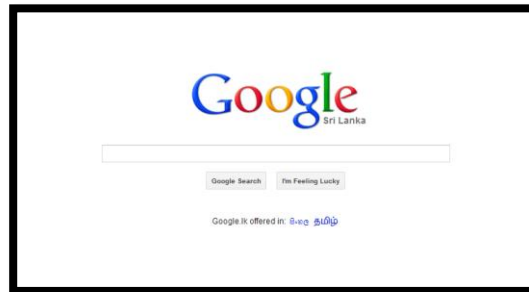
Actual Result - Pass



Test Case : Admin Login**Pre Conditions : The admin wants to login to the system**

TC #	Sequence #	Test Description	Input Value	Expected Result
1	1.1	Correct user name and correct password	Admin "Admin123"	Welcome pages displayed - Pass
	1.2	Correct user name and incorrect password	Admin 123	Error should be displayed - Pass

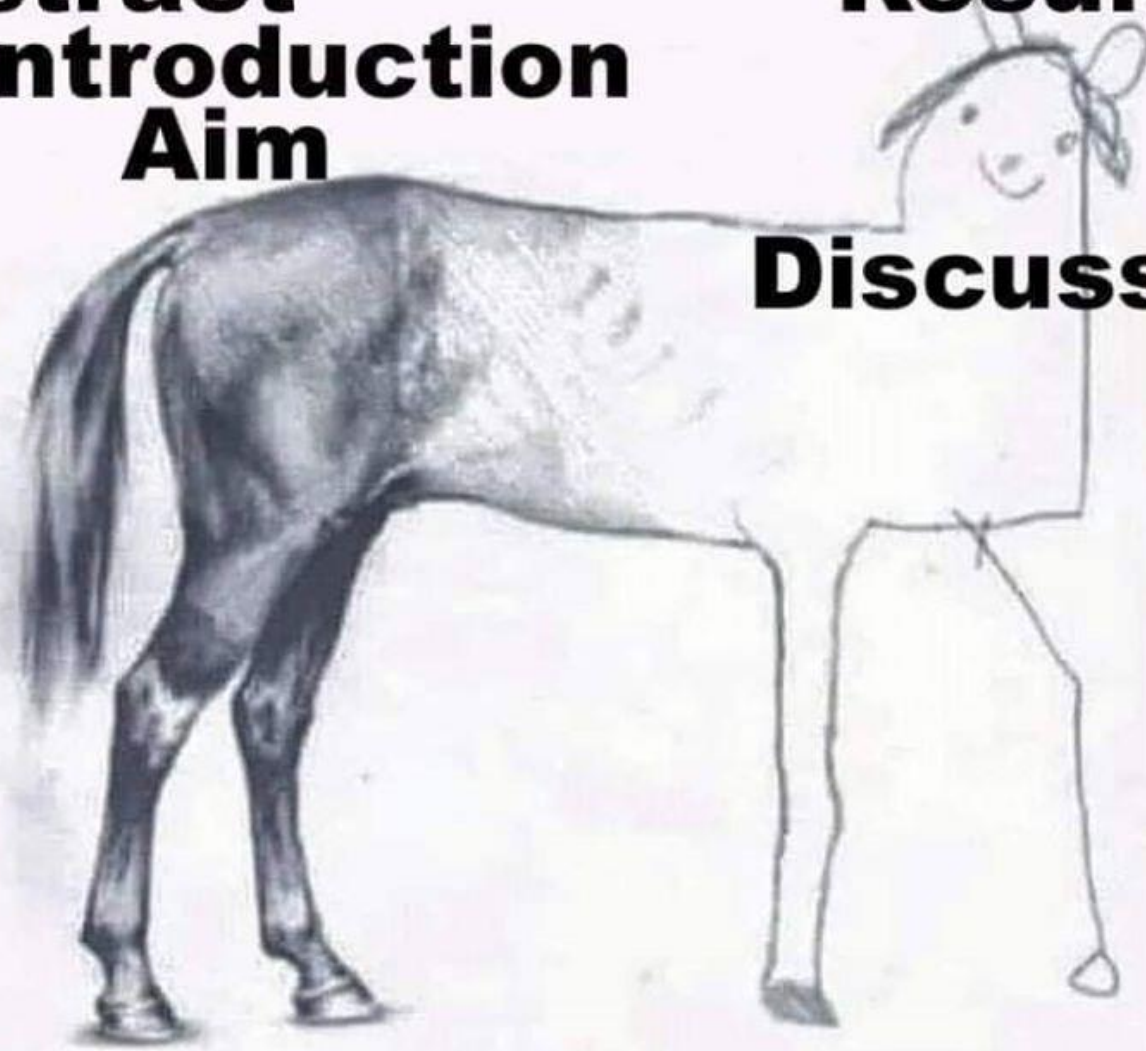
Actual Result - Pass



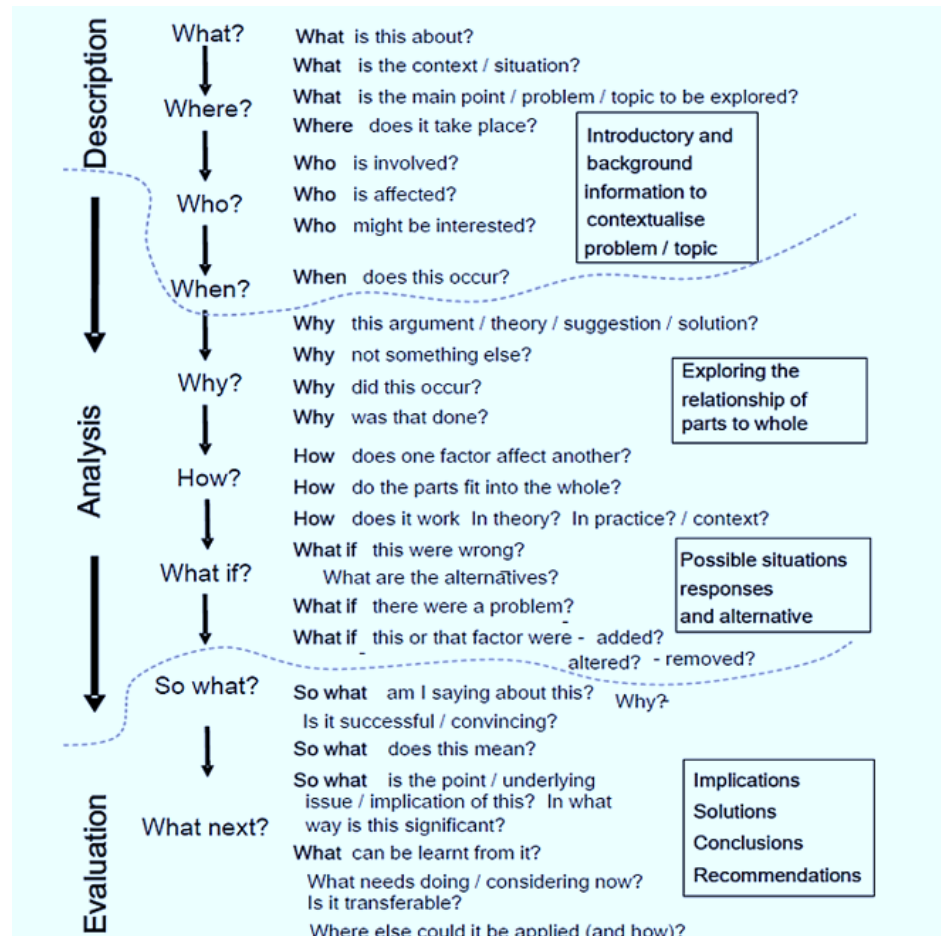
Abstract
Introduction
Aim

Results

Discussion



Analysis and evaluation - showing your critical thinking



Discussion

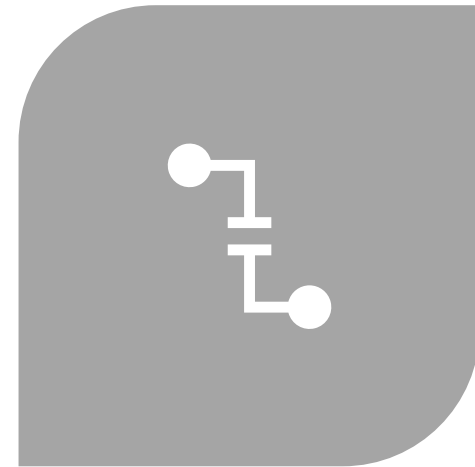
Discussion

- Reliability / Accuracy level of your system
- Reasons for those levels
- Problems faced (Technical)
- How you solved them
- Any reason get deviated from the original plans

Evaluation



HOW WAS USER FEEDBACK
COLLECTED ?



MODIFICATIONS AND FINE-
TUNING POINTS



What does the 'Evaluation' section do?

- It links your testing/feedback results to the original aims and objectives
- It focuses mainly on the artefact strengths
- It describes your testing results and feedback
- It says how the design could have been or could be improved
- It gives your judgement on how successful your project has been



What does the 'Evaluation' section do?

- It links your testing/feedback results to the original aims and objectives
- ~~• It focuses mainly on the artefact strengths~~
- ~~• It describes your testing results and feedback~~
- It says how the design could have been or could be improved
- It gives your judgement on how successful your project has been

Evaluation Creates the link

This student needs to find out

- Did it meet the specifications?
- Was the artefact relevant?
- Did users enjoy it?
- What was my supervisor's standard and did I meet it?
- Did users want to use it more often?

Answer these evaluation questions

Consider implications, solutions, conclusions and recommendations.

This involves asking **‘So what?’** and **‘What next?’** type questions. Such as:

- What do these results mean?
- Why are they significant?
- Are they convincing? Why or why not?
- What are the implications?

- Is my project successful? Why/why not?
- How does it meet the criteria?
- Is it transferable? Why/why not?
- How and where else can it be applied?
- What can be learnt from it?
- What needs doing now (future work)?

(Study Hub, nd)

Who says
thesis
writing is
difficult?

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

Summary

Don't be lazy, just read chapter 1 to 4.

Conclusion

Within the data gathered, the researchers just want to submit the research to find some peace.

Recommendation

The researchers highly recommend Mouse, Vagabond, Queen of Tears, Lovely Runner, When the Phone Rings and 99 other kdramas.

Conclusion

Importance of outcome

How were objectives met?

Limitations

Recommendations

Future work

References – Harvard Style

**Me in school
wondering why
plagiarism is
such a big deal**



**Me when
someone steals
my memes
without credit**



Appendices

A4 size project poster

Questionnaires

Results

Research paper

Additional interfaces

Additional code

Reference

- Presentation slides – Final report writing (CIS017-3) by Liz Hulford Wood (Learning Development tutor) / Janine Bhandol (Librarian) - UoB

Thank you

